

ISAIAH KOL

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SUMMARY

Mechanical engineering student with a background in competitive robotics and hands-on experience designing and building mechanical systems through CAD, fabrication, and iterative prototyping.

EDUCATION

University of Vermont | Burlington, VT Aug 2024 – Present
Bachelor of Science, Mechanical Engineering | **Expected May 2028**
- GPA: 3.80 | Dean's List

EXPERIENCE

UVM FabLab | Burlington, VT Dec 2025 – Present
Operations Manager
- Drive workflow, equipment, and space improvements to optimize lab efficiency and engagement.
- Coordinate technicians and oversee day-to-day lab and equipment operations.
- Balance leadership duties with full technician responsibilities.
Technician Sep 2025 – Dec 2025
- Operated 15+ 3D printers and other equipment for student projects and university research.
- Assisted students and researchers with CAD, projects, and in-house equipment use.
- Supported lab organization, safety practices, and routine machine maintenance.

Dartmouth College Dining Services Line Cook & Counterworker | Hanover, NH Aug 2022 – Aug 2024
- Balanced customer service with food prep and cooking in a fast-paced dining environment.

Gap-Year: VEX Robotics Team Coach | Thetford, VT Aug 2023 – Jun 2024
- Mentored students aged 12-18 in CAD, design process, safe fabrication, and competition strategy.

VEX Robotics Team 4886-B | Various Locations, USA Sep 2017 – Jun 2023
- Collaboratively responsible for robot design, construction, controls, and technical documentation.
- Four-time competitors at VEX World Championship; 2023 NH/VT State Champions; recipients of three Design Awards and three Excellence Awards at the regional (NH/VT) level.

Internship at Hypertherm & Fujifilm Dimatix | Lebanon, NH Aug 2021 – Jan 2022
- Gained exposure to manufacturing practices and engineering environments.
- Built foundational skills in CAD and CNC/manual machining through guided projects.

PROJECTS

Cycloidal Drive
- Designed and fabricated a 22:1 reduction cycloidal gearbox for a NEMA-17 stepper motor.
- Iterated cycloidal geometry and FDM printing tolerances to minimize backlash.
- Achieved highly compact form by matching axial gearbox length to motor shaft length (20 mm).
- Tested actuation with an Arduino UNO as the first part of an in-progress robot arm project.

Filament Recycler Spool Holder
- Identified constraint in FabLab's PLA recycling system regarding brand-specific spool holder geometry.
- Designed and 3D printed an alternative fixture enabling the use of any generic spool on the machine.
- Improved filament recycling workflow, allowing use of the lab's empty spool stockpile.

SKILLS

Design: SOLIDWORKS, AutoCAD, Design for Manufacturability (DFM), Technical Documentation & Communication
Fabrication: FDM 3D Printing, Machining Basics, Automotive Repair
Software: Excel, PowerPoint, Python Basics, FDM Slicing Software